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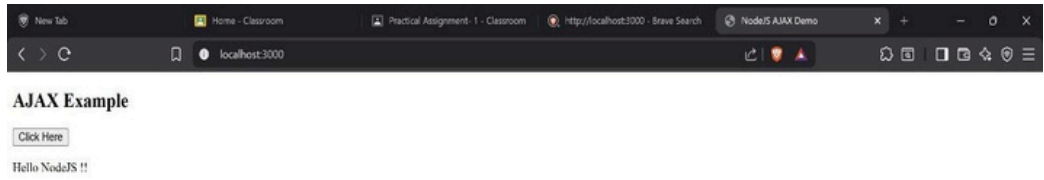
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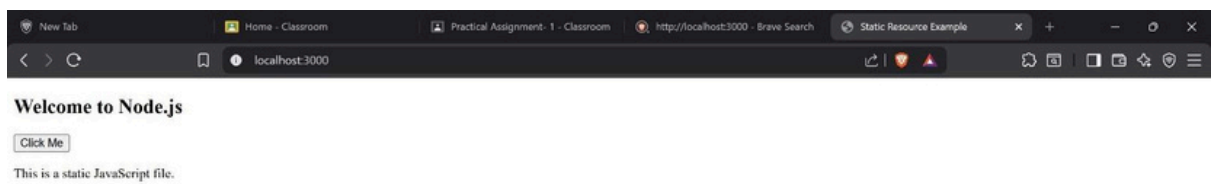
https://github.com/2023040284-ssp/56_SurajPatel_701_A1

ProblemSheet - 1 – Solution

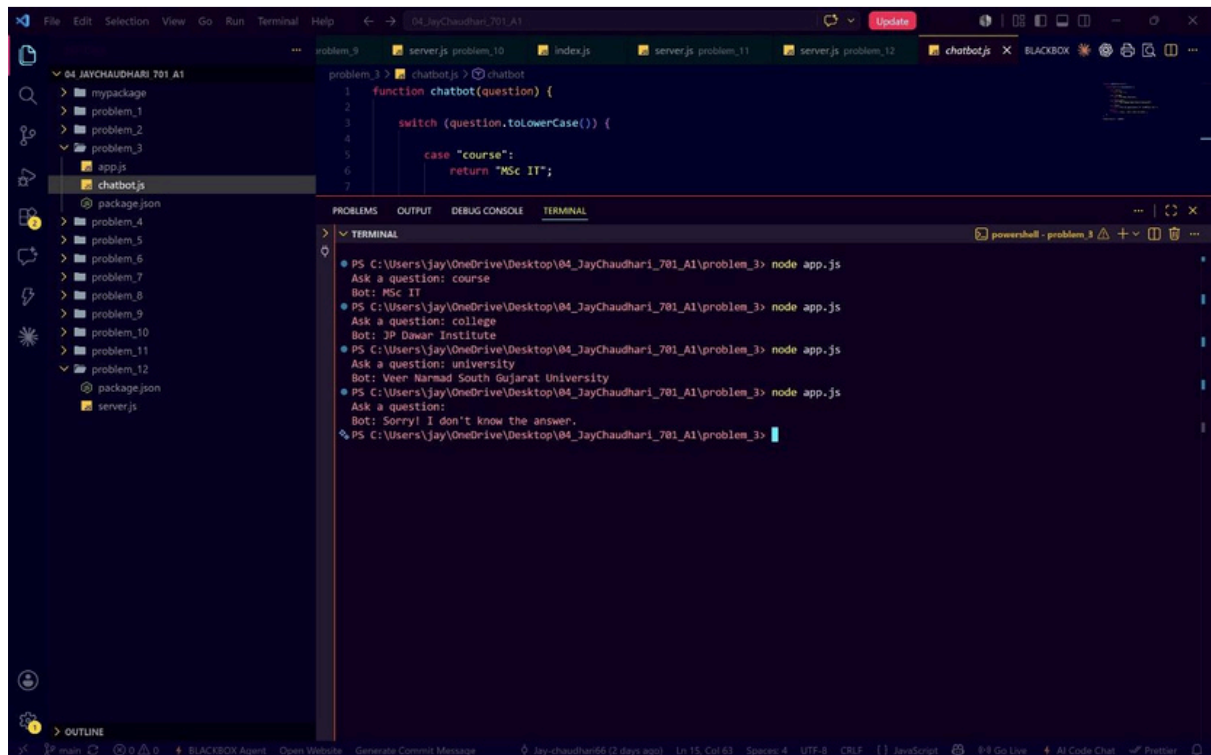
Q1 . output:



Q2 . output:



Q3 . output:



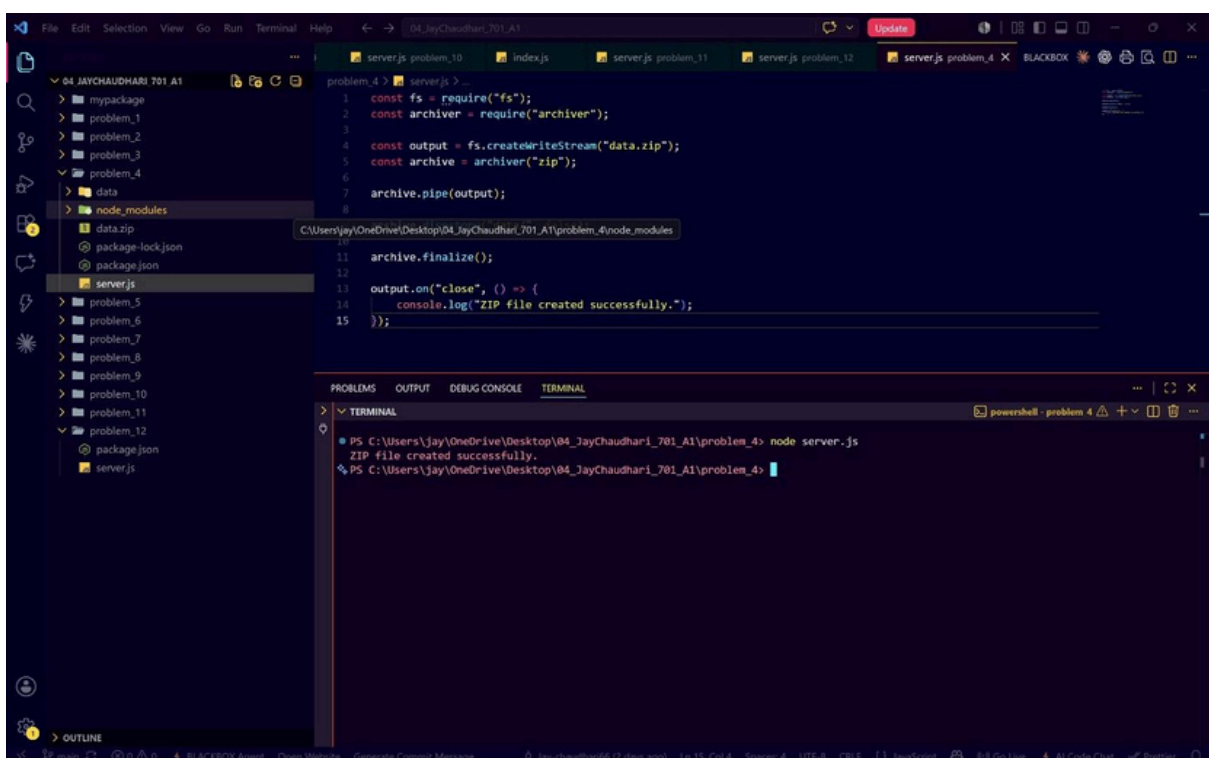
The screenshot shows a VS Code editor window with a file explorer on the left and a terminal at the bottom. The file explorer shows a project named '04_JAYCHAUDHARI_701_A1' with a folder 'chatbotjs' containing 'app.js'. The 'chatbotjs' folder is selected. The 'app.js' file is open in the editor, showing a function 'chatbot(question)' that uses a switch statement to return different responses based on the input question. The terminal shows the output of running 'node app.js' for three different questions: 'course', 'college', and 'university'. The responses are 'Msc IT', 'JP Dawar Institute', and 'Veer Narmad South Gujarat University' respectively. For an empty input, the response is 'Sorry! I don't know the answer.'.

```
function chatbot(question) {  
  switch (question.toLowerCase()) {  
    case "course":  
      return "Msc IT";  
  }  
}
```

Terminal Output:

```
PS C:\Users\jay\OneDrive\Desktop\04_JayChaudhari_701_A1\problem_3> node app.js  
Ask a question: course  
Bot: Msc IT  
PS C:\Users\jay\OneDrive\Desktop\04_JayChaudhari_701_A1\problem_3> node app.js  
Ask a question: college  
Bot: JP Dawar Institute  
PS C:\Users\jay\OneDrive\Desktop\04_JayChaudhari_701_A1\problem_3> node app.js  
Ask a question: university  
Bot: Veer Narmad South Gujarat University  
PS C:\Users\jay\OneDrive\Desktop\04_JayChaudhari_701_A1\problem_3> node app.js  
Ask a question:  
Bot: Sorry! I don't know the answer.  
PS C:\Users\jay\OneDrive\Desktop\04_JayChaudhari_701_A1\problem_3>
```

Q4 . output :



The screenshot shows a VS Code editor window with a file explorer on the left and a terminal at the bottom. The file explorer shows a project named '04_JAYCHAUDHARI_701_A1' with a folder 'serverjs' containing 'server.js'. The 'serverjs' folder is selected. The 'server.js' file is open in the editor, showing a script that creates a ZIP file named 'data.zip' using the 'fs' and 'archiver' modules. The terminal shows the output of running 'node server.js', which is 'ZIP file created successfully.'.

```
const fs = require("fs");  
const archiver = require("archiver");  
const output = fs.createWriteStream("data.zip");  
const archive = archiver("zip");  
archive.pipe(output);  
archive.finalize();  
output.on("close", () => {  
  console.log("ZIP file created successfully.");  
});
```

Terminal Output:

```
PS C:\Users\jay\OneDrive\Desktop\04_JayChaudhari_701_A1\problem_4> node server.js  
ZIP file created successfully.  
PS C:\Users\jay\OneDrive\Desktop\04_JayChaudhari_701_A1\problem_4>
```

Q5 . output:

The screenshot shows the VS Code editor with a project named '04_JAYCHAUDHARI_701_A1'. The file explorer on the left shows a directory structure with 'problem_4' selected. The editor displays the code for 'problem_4.js', which uses the 'fs' and 'archiver' modules to create a ZIP file named 'data.zip'. The code includes comments and console logs for feedback. The terminal at the bottom shows the execution of 'node server.js' in the 'problem_4' directory, resulting in the message 'ZIP file created successfully.'.

```
1 const fs = require("fs");
2 const archiver = require("archiver");
3
4 const output = fs.createWriteStream("data.zip");
5 const archive = archiver("zip");
6
7 archive.pipe(output);
8
9 archive.directory("data/", false);
10
11 archive.finalize();
12
13 output.on("close", () => {
14   console.log("ZIP file created successfully.");
15 });
```

TERMINAL

```
PS C:\Users\Jay\OneDrive\Desktop\04_JayChaudhari_701_A1\problem_4> node server.js
ZIP file created successfully.
PS C:\Users\Jay\OneDrive\Desktop\04_JayChaudhari_701_A1\problem_4> cd ..
PS C:\Users\Jay\OneDrive\Desktop\04_JayChaudhari_701_A1> cd problem_5
PS C:\Users\Jay\OneDrive\Desktop\04_JayChaudhari_701_A1\problem_5> node server.js
ZIP file extracted successfully.
PS C:\Users\Jay\OneDrive\Desktop\04_JayChaudhari_701_A1\problem_5>
```

Q6 . output :

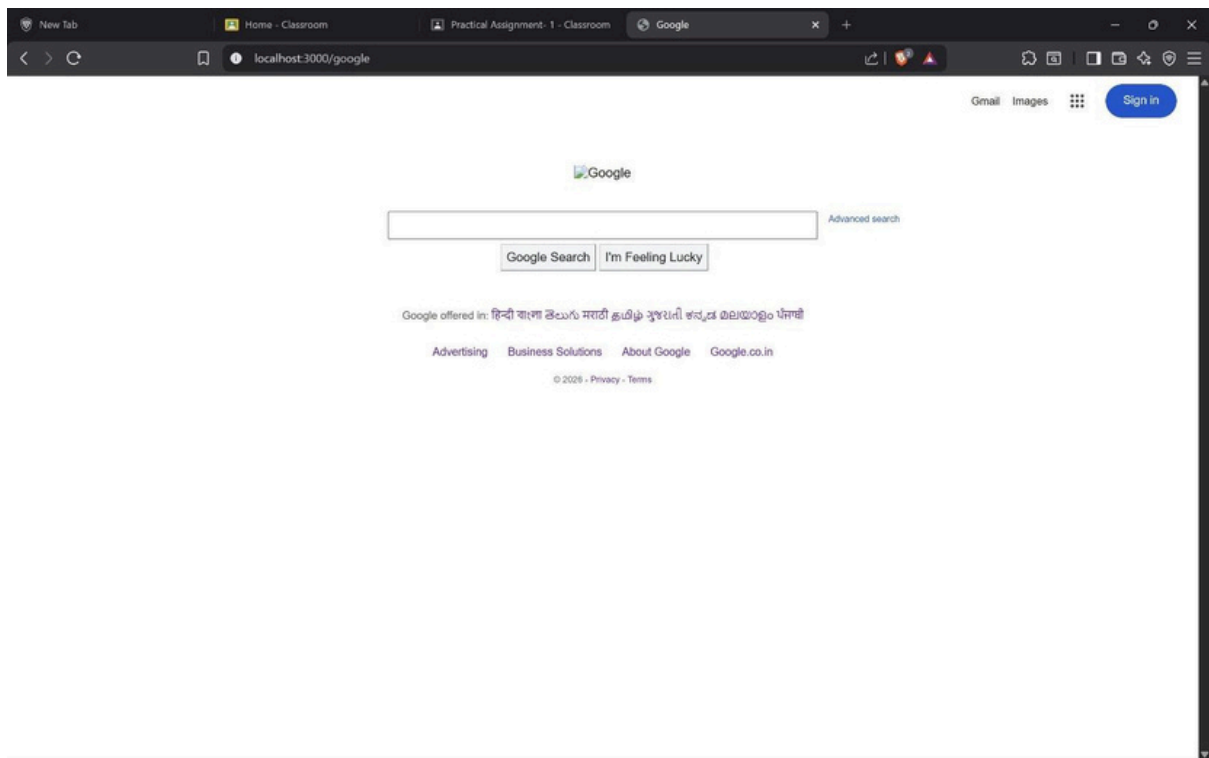
The screenshot shows the VS Code editor with the same project. The file explorer shows 'problem_6' selected. The editor displays the code for 'problem_6.js', which uses the 'fs' and 'util' modules to delete a file named 'sample.txt'. The code includes comments and console logs for feedback. The terminal at the bottom shows the execution of 'node server.js' in the 'problem_6' directory, resulting in the message 'File deleted successfully.'.

```
1 const fs = require("fs");
2 const util = require("util");
3
4 const unlinkFile = util.promisify(fs.unlink);
5
6 unlinkFile("sample.txt")
7   .then(() => {
8     console.log("File deleted successfully.");
9   })
10  .catch(err => {
11    console.log(err);
12  });
```

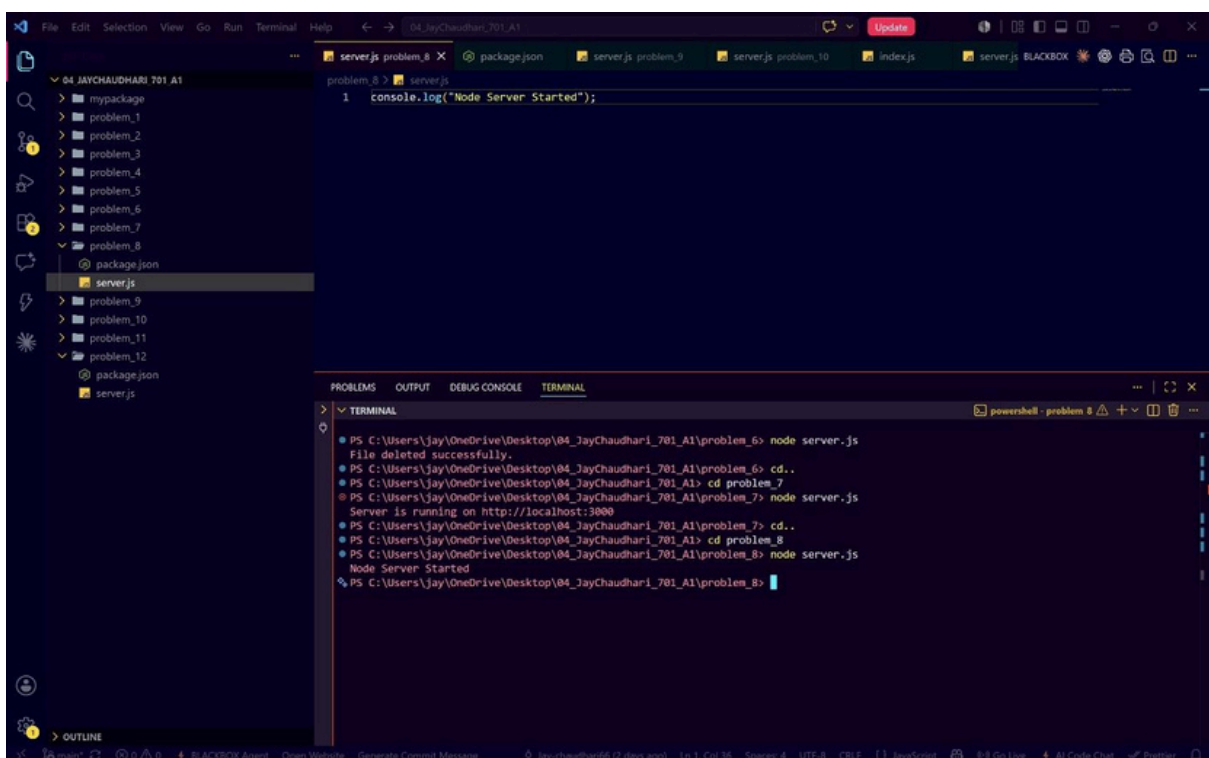
TERMINAL

```
PS C:\Users\Jay\OneDrive\Desktop\04_JayChaudhari_701_A1\problem_6> node server.js
File deleted successfully.
PS C:\Users\Jay\OneDrive\Desktop\04_JayChaudhari_701_A1\problem_6>
```

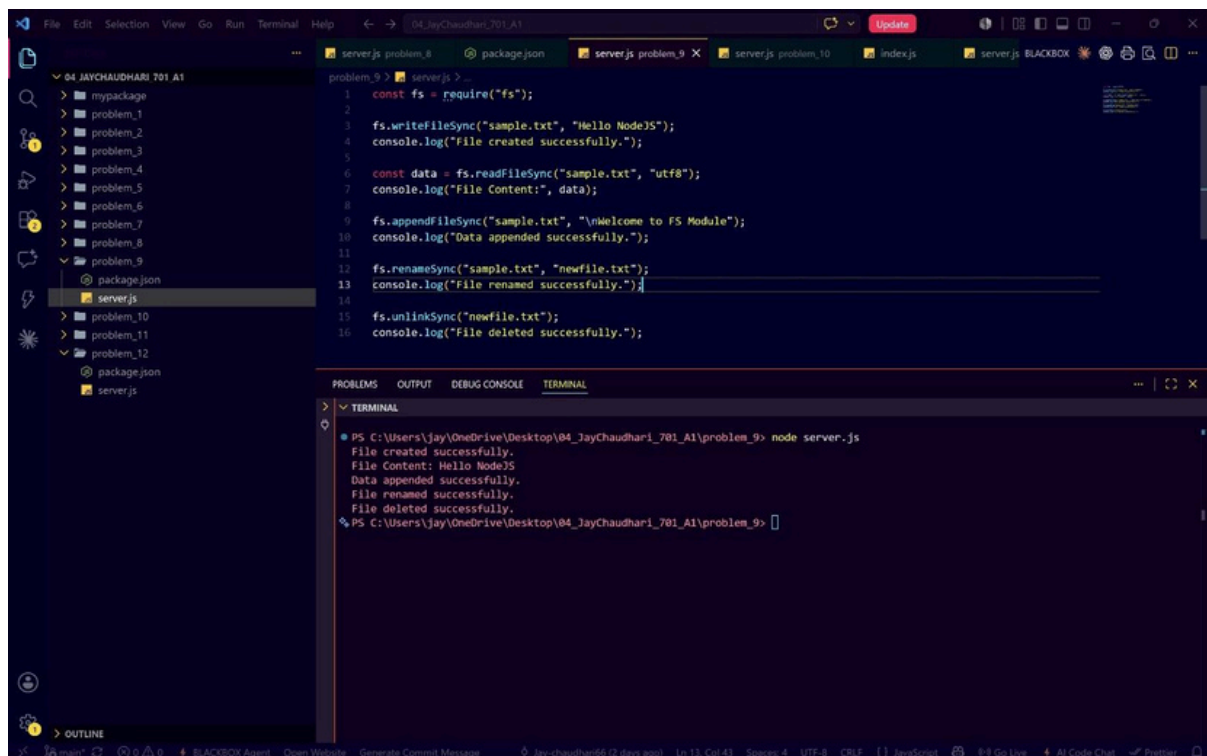
Q7 . output :



Q8 . output:



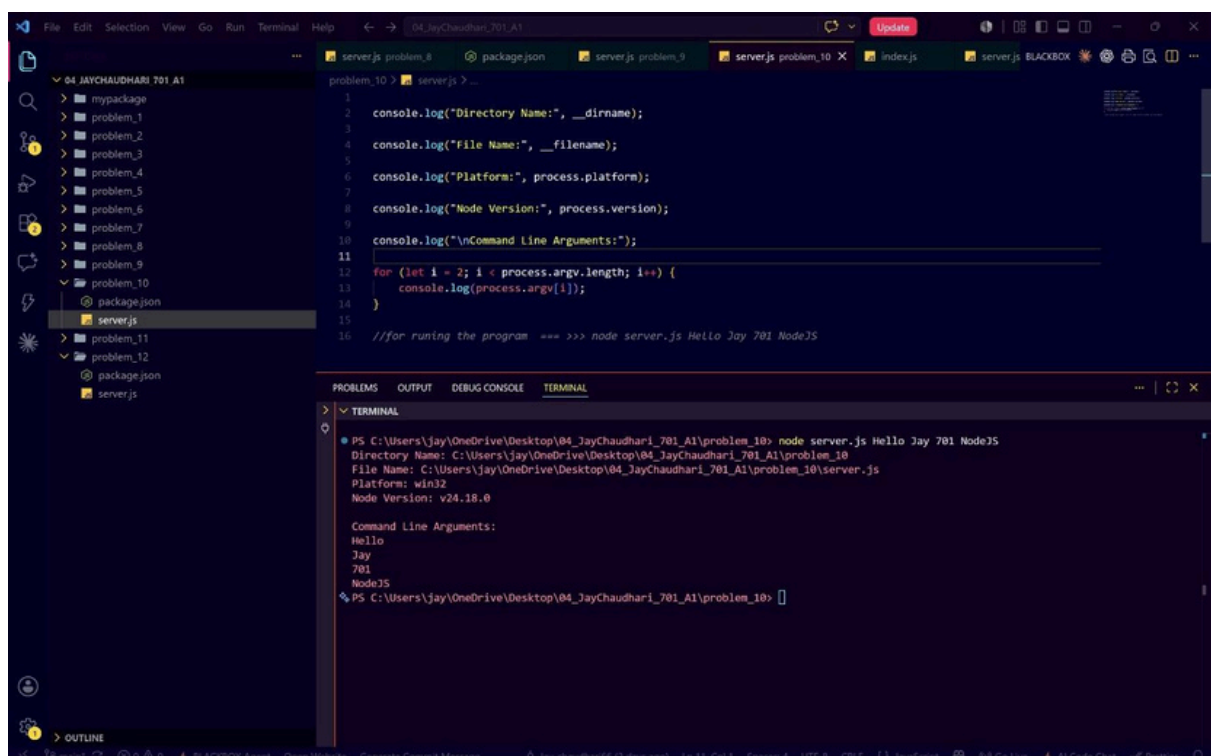
Q9 . output:



The screenshot shows a VS Code window with a file explorer on the left displaying a directory structure for '04_JAYCHAUDHARI_701_A1'. The main editor shows the code for 'server.js' in 'problem_9'. The code uses the 'fs' module to create, read, append, rename, and delete files. The terminal at the bottom shows the command 'node server.js' being executed, with the following output:

```
PS C:\Users\Jay\OneDrive\Desktop\04_JayChaudhari_701_A1\problem_9> node server.js
File created successfully.
File Content: Hello NodeJS
Data appended successfully.
File renamed successfully.
File deleted successfully.
```

Q10 . output:

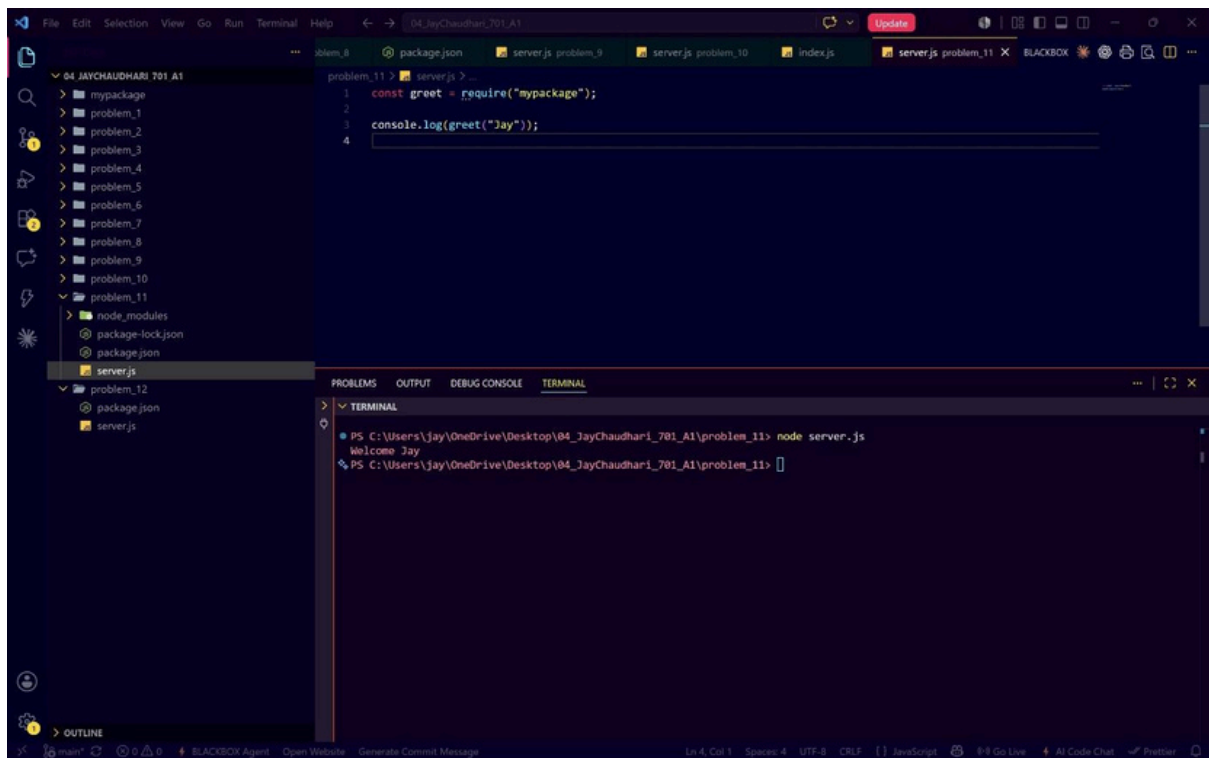


The screenshot shows a VS Code window with a file explorer on the left displaying a directory structure for '04_JAYCHAUDHARI_701_A1'. The main editor shows the code for 'server.js' in 'problem_10'. The code uses 'console.log' to display system information and command line arguments. The terminal at the bottom shows the command 'node server.js Hello Jay 701 NodeJS' being executed, with the following output:

```
PS C:\Users\Jay\OneDrive\Desktop\04_JayChaudhari_701_A1\problem_10> node server.js Hello Jay 701 NodeJS
Directory Name: C:\Users\Jay\OneDrive\Desktop\04_JayChaudhari_701_A1\problem_10
File Name: C:\Users\Jay\OneDrive\Desktop\04_JayChaudhari_701_A1\problem_10\server.js
Platform: win32
Node Version: v24.18.0

Command Line Arguments:
Hello
Jay
701
NodeJS
```

Q11 . output:



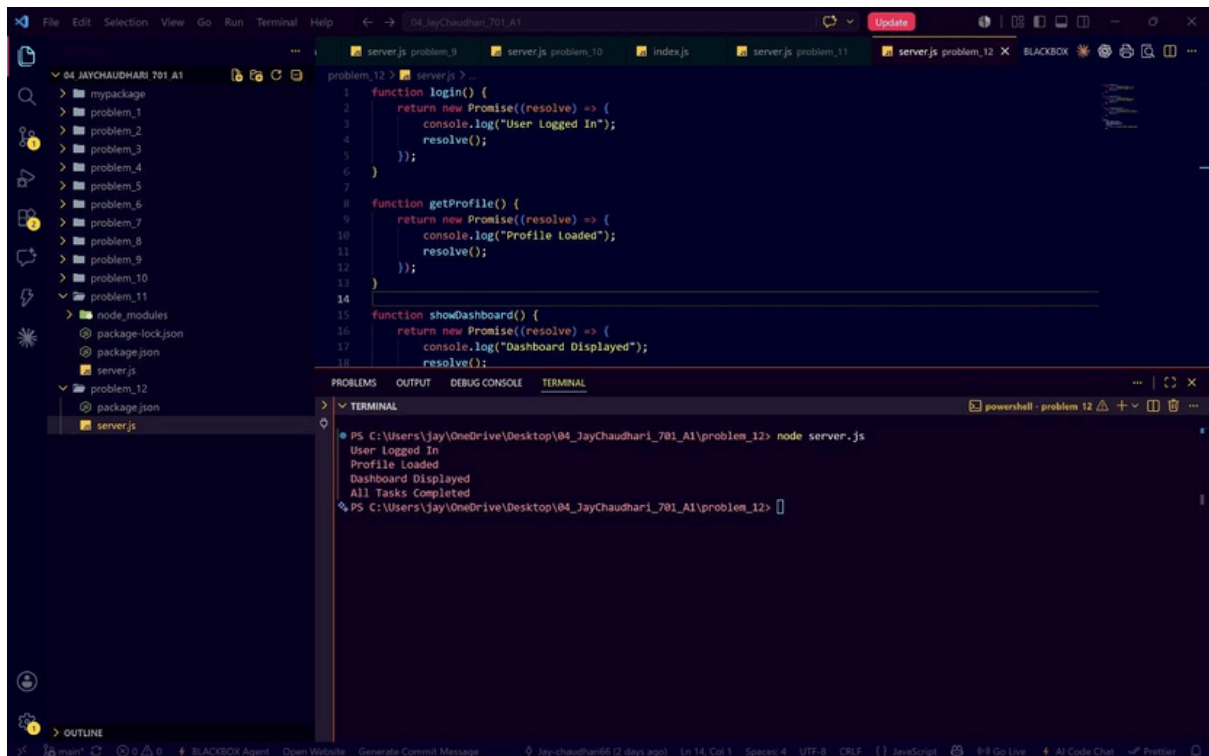
The screenshot shows the VS Code interface with a project named '04_JAYCHAUDHARI_701_A1'. The file explorer on the left shows a directory structure with 'mypackage', 'problem_1' through 'problem_11', 'node_modules', 'package-lock.json', and 'package.json'. The 'server.js' file is selected in the 'problem_11' folder. The editor displays the following code:

```
1 const greet = require("mypackage");
2
3 console.log(greet("Jay"));
4
```

The terminal at the bottom shows the command 'node server.js' being executed, resulting in the output 'Welcome Jay'.

```
PS C:\Users\jay\OneDrive\Desktop\04_JayChaudhari_701_A1\problem_11> node server.js
Welcome Jay
PS C:\Users\jay\OneDrive\Desktop\04_JayChaudhari_701_A1\problem_11>
```

Q12 . output :



The screenshot shows the VS Code interface with the same project. The file explorer shows the 'server.js' file selected in the 'problem_12' folder. The editor displays the following code:

```
1 function login() {
2   return new Promise((resolve) => {
3     console.log("User Logged In");
4     resolve();
5   });
6 }
7
8 function getProfile() {
9   return new Promise((resolve) => {
10    console.log("Profile Loaded");
11    resolve();
12  });
13 }
14
15 function showDashboard() {
16   return new Promise((resolve) => {
17     console.log("Dashboard Displayed");
18     resolve();
19   });
20 }
21
```

The terminal at the bottom shows the command 'node server.js' being executed, resulting in the output 'User Logged In', 'Profile Loaded', 'Dashboard Displayed', and 'All Tasks Completed'.

```
PS C:\Users\jay\OneDrive\Desktop\04_JayChaudhari_701_A1\problem_12> node server.js
User Logged In
Profile Loaded
Dashboard Displayed
All Tasks Completed
PS C:\Users\jay\OneDrive\Desktop\04_JayChaudhari_701_A1\problem_12>
```